ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Temperature and its Measurement** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 50 °C	Linear scale with equal spacing gives 50 °C.
2	**(b)** 40 s	Twice slower.
3	**(b)** aluminium and plastic	Pan conducts, handle insulates.
4	**(d)** metal cup	Metal conducts heat away fastest.
5	**(b)** the 20 °C object	Heat enters the colder one.
6	**(b)** metal rod hotter	Metal conducts quickly.
7	**(d)** heat in equals heat out	Balance.
8	**(d)** normal temperature with small variation	Normal range.
9	**(a)** Both A and R are true and R is the correct explanation of A.	After holding a hand in cold water, lukewarm water feels warm; this shows touch can mislead, which R explains.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but the two uses are separate and R does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Temperatures are equal (1); The larger quantity contains more heat energy (1).	1+1
12	Alcohol does not freeze easily (1); It is safer (1).	1+1
13	The bucket can have more heat (1); Temperature shows hotness, not amount (1).	1+1
14	No, both are at room temperature (1); Metal conducts heat from the hand faster (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Atmospheric pressure is lower on hills (1); Water boils at a lower temperature (1); Therefore reading differs (1).	1+1+1
16	Reduces loss of hot air (1); Stops steam escaping (1); Keeps heat (1).	1+1+1
17	Touching the bottom (1); Reading from an angle (1); Not waiting for a steady reading (1).	1+1+1
	OR: Same amount of water at same temperature (1); Wrap each beaker (1); Measure after 30 minutes (1).	1+1+1
18	Record patterns (1); Notice rise or fall (1); Use trends for prediction (1).	1+1+1
	OR: Hot and cold bowl for each hand (1); Then both in the middle bowl (1); Feelings conflict (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) To bring the mercury thread down so that the old reading does not mislead (1). (ii) About 37 degrees C (1). (iii) Degrees Celsius (1). (iv) Yes, 38.5 degrees C is higher than 37 degrees C (1); a hot-water temperature is beyond its range and the thermometer would break (1).	1+1+1+2
20	Particles at the hot end gain energy (1); Pass on to neighbours (1); Heat travels to the cooler end (1); Plastic is an insulator (1); Slow transfer (1).	1+1+1+1+1
	OR: Bulb touching the vessel (1); Reading at an angle (1); Not waiting (1); Wrong thermometer (1); Follow the steps carefully (1).	1+1+1+1+1

Mistakes to avoid:

- Using a clinical thermometer for hot water or for a room; it is only for the **human body**.
- Not shaking the clinical thermometer before use, so an old reading is mistaken for the new one.
- Letting the bulb of the laboratory thermometer touch the beaker, or reading it from a slant.
- Believing that our hands can measure temperature accurately; touch can be **misleading**.
- Writing that cold flows from ice to a hand; **heat** flows from the hotter to the colder object.
- Writing the wrong units or values: body temperature is about 37 degrees C, not 100 degrees C.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Temperature and its Measurement** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 20 °C	Below its range.
2	**(a)** how hot something is	Hotness.
3	**(b)** a bigger object is not always hotter	Temperature is independent of size.
4	**(a)** the one in metal foil	Wool insulates.
5	**(b)** ice, room, body, boiling water	0, ~25, 37, 100.
6	**(b)** boiling water	Temperature is about hotness.
7	**(d)** thermocol box	Insulator.
8	**(b)** the temperature difference with the room reduces	Rate depends on difference.
9	**(d)** A is false but R is true.	A is false: the range is too small and it would break. R is true.
10	**(a)** Both A and R are true and R is the correct explanation of A.	The tea is hotter than the air, so heat passes from tea to the surroundings.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The mercury must reach the body temperature (1); An early reading would be too low (1).	1+1
12	0 °C and 100 °C are reference marks (1); Ensure accurate scale (1).	1+1
13	It changes to steam (1); Heat is used for the change of state (1).	1+1
14	Sudden change can crack the glass (1); Gives a wrong reading (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Larger temperature difference at first (1); Rate of heat loss reduces (1); Approaches room temperature slowly (1).	1+1+1
16	At 15 °C they take heat from the hand (1); At 50 °C they give heat to hand (1); Metal conducts quickly in both (1).	1+1+1
17	Cold surface cools air (1); Water vapour condenses (1); Droplets form (1).	1+1+1
	OR: Clinical has a limited range (1); Mercury would burst the bulb (1); Use a laboratory thermometer (1).	1+1+1
18	Iron cools (1); Water warms (1); Both reach a common temperature (1).	1+1+1
	OR: Snow traps air (1); Air insulates (1); Heat from ground is retained (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Laboratory thermometer (1). (ii) The beaker wall may be hotter or colder than the water and give a wrong reading (1). (iii) Level with the mercury thread (1). (iv) Heat has flowed from the hot water to the cooler surroundings (1), so the water got cooler (1).	1+1+1+2
20	Fall 20 in the first 5 min (1); Fall 12 in the next (1); Fall 6 in the next (1); Rate of fall decreases (1); Approaches room temperature (1).	1+1+1+1+1
	OR: Iron at 20 °C feels cold (1); Wood at 20 °C feels warmer (1); Both same temperature (1); Iron removes heat faster (1); Touch senses heat loss (1).	1+1+1+1+1

Mistakes to avoid:

- Using a clinical thermometer for hot water or for a room; it is only for the **human body**.
- Not shaking the clinical thermometer before use, so an old reading is mistaken for the new one.
- Letting the bulb of the laboratory thermometer touch the beaker, or reading it from a slant.
- Believing that our hands can measure temperature accurately; touch can be **misleading**.
- Writing that cold flows from ice to a hand; **heat** flows from the hotter to the colder object.
- Writing the wrong units or values: body temperature is about 37 degrees C, not 100 degrees C.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Temperature and its Measurement** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** to resist heat conduction	Insulation.
2	**(d)** pour it into a wide steel plate	Metal and large surface lose heat faster.
3	**(a)** a small expansion makes a large change in length	Better sensitivity.
4	**(c)** metal conducts heat away from the hand	Conduction rate.
5	**(b)** body parts differ in temperature	Armpit reading is lower.
6	**(a)** small changes in fever matter	Precision for body temperature.
7	**(b)** air gets trapped and heat loss is less	Less convection.
8	**(c)** touching with hand	Touch is unreliable.
9	**(c)** A is true but R is false.	A is true; R is false, the normal value is about 37 degrees C.
10	**(a)** Both A and R are true and R is the correct explanation of A.	A kink stops the mercury going back, so the thread must be shaken down first.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Air in between acts as insulation (1); Air is a poor conductor (1).	1+1
12	Slightly above 37 °C (1); Mild fever (1).	1+1
13	Kink stops fall when removed (1); Shaking forces mercury through the kink (1).	1+1
14	Steel conducts heat away (1); Insulated tiffin is better (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Ice is at lower temperature (1); Heat flows from drink to ice (1); Ice melts and the drink cools (1).	1+1+1
16	Keep the inside cold (1); Stops heat entering (1); Saves energy (1).	1+1+1
17	Body or base conducts heat (1); Handle insulates (1); Both needed (1).	1+1+1
	OR: Bulb receives heat from water (1); Mercury expands (1); Becomes steady at water's temperature (1).	1+1+1
18	Glass may break (1); Mercury is harmful (1); Digital is safer (1).	1+1+1
	OR: Mark the level in ice as 0 (1); Mark the level in boiling water as 100 (1); Divide into equal parts (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Temperature (1). (ii) Thermometer (1). (iii) The sense of touch (1). (iv) Each hand had got used to a different temperature (1), so touch gave misleading feelings and cannot be trusted to measure temperature (1).	1+1+1+2
20	A spark is very hot but has little heat (1); A bucket of warm water has more heat (1); Ocean vs cup (1); Temperature shows hotness (1); Heat depends on quantity (1).	1+1+1+1+1
	OR: Cooking at right temperature kills germs (1); Fridge slows spoilage (1); Do not leave food warm for hours (1); Check with thermometer (1); Reheat well (1).	1+1+1+1+1

Mistakes to avoid:

- Using a clinical thermometer for hot water or for a room; it is only for the **human body**.
- Not shaking the clinical thermometer before use, so an old reading is mistaken for the new one.
- Letting the bulb of the laboratory thermometer touch the beaker, or reading it from a slant.
- Believing that our hands can measure temperature accurately; touch can be **misleading**.
- Writing that cold flows from ice to a hand; **heat** flows from the hotter to the colder object.
- Writing the wrong units or values: body temperature is about 37 degrees C, not 100 degrees C.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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